SUDARSHAN AI
User & Installation Manual
Explainable Deepfake and Video Manipulation Detection System

Version: 1.0
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Research Group: Vision and Signal Information Processing Research Group
Supported Platforms: Windows, macOS, Linux

1. Introduction

About SUDARSHAN AI

SUDARSHAN AI is an advanced explainable Artificial Intelligence system designed for the detection of deepfakes and manipulated visual media. The platform combines multiple analysis methodologies to identify synthetic content and provide interpretable results for researchers, academic institutions, organizations, and investigators.

The system employs a multi-modal detection framework that integrates:

- Frequency Domain Analysis
- Temporal Consistency Analysis
- Spatial Feature Analysis

By combining these independent detection streams, SUDARSHAN AI provides robust deepfake detection capabilities together with transparent and explainable outputs for both technical and non-technical users.

2. System Requirements

Minimum Hardware Requirements

Windows
- Windows 10 or later
- Intel Core i5 / AMD Ryzen 5 or higher
- 8 GB RAM (16 GB recommended)
- 5 GB free storage space

macOS

- macOS 12 Monterey or later
- Apple Silicon (M-Series) or Intel Processor
- 8 GB RAM minimum
- 5 GB free storage space

Linux
- Ubuntu 20.04+, Debian-based distributions, or equivalent
- 8 GB RAM minimum
- 5 GB free storage space

Software Requirements

Python Installation

Python Version 3.10 or higher is required.

Windows
Download and install Python from:

https://www.python.org/downloads/windows/

Important: During installation, ensure that the option:

"Add python.exe to PATH"

is selected before proceeding.

macOS

Download Python from:

https://www.python.org/downloads/macos/

or install using Homebrew:

bash brew install python

Linux

Install Python using:

bash sudo apt update sudo apt install python3

3. Platform-Specific Dependencies

Windows

No additional dependencies are required.

macOS

If Tkinter is not available, install:

bash brew install python-tk

Linux

Install required packages:

bash sudo apt-get update sudo apt-get install python3-tk python3-venv

4. Installation Guide

Windows Installation

1. Download the SUDARSHAN AI package.
2. Extract the ZIP archive.
3. Open the extracted folder.
4. Double-click:

text Install_SUDARSHAN.bat

5. The installer will:
   - Verify Python installation
   - Create a virtual environment
   - Install dependencies
   - Configure the application

6. Wait until installation completes.

macOS Installation

1. Download:

text SUDARSHAN_AI_Mac_v1.0.dmg

2. Open the DMG file.
3. Drag the SUDARSHAN AI application into the Applications folder.
4. Eject the mounted image.
5. Launch the application from Applications.

During the first launch, the software may automatically configure dependencies.

Linux Installation

1. Extract the package.
2. Open Terminal inside the extracted directory.
3. Run:

bash chmod +x install_linux.sh chmod +x launch_linux.sh ./install_linux.sh

4. The installation utility will:
   - Verify dependencies
   - Create the environment
   - Install required packages
   - Validate installation

5. Launching SUDARSHAN AI

Windows

Double-click:

text Launch_SUDARSHAN.bat

macOS

Open:

text Applications → SUDARSHAN AI

or search using Spotlight.

Linux

Run:

bash ./launch_linux.sh

6. User Interface Overview

The application interface consists of:

- Input Media Panel
- Analysis Controls
- Detection Results Panel
- Explainability Reports
- Visual Heatmap Viewer
- Live Monitoring Dashboard

7. Operating Modes

Mode 1: Video Analysis

Purpose

Analyze prerecorded videos for deepfake indicators.

Supported Formats

- MP4
- AVI

- MOV

Procedure

1. Select "Upload Video".
2. Choose the desired video file.
3. Start analysis.
4. Monitor progress indicators.
5. Review:
   - Detection confidence
   - Explainability reports
   - Visual heatmaps

Output

- Deepfake probability score
- Detection classification
- Spatial attention maps
- User explanation report
- Developer explanation report

Mode 2: Static Image Analysis

Purpose

Analyze individual images for manipulation indicators.

Supported Formats

- JPG
- JPEG
- PNG
- WEBP

Procedure

1. Click "Upload Image".
2. Select an image.
3. Start analysis.
4. Review generated results.

Note

Temporal analysis is not applicable to static images.

Mode 3: Live Webcam Monitoring

Purpose

Real-time deepfake monitoring through webcam input.

Procedure

1. Click "Start Live Stream".
2. Allow camera permissions if prompted.
3. Real-time inference begins automatically.
4. Monitor detection results.
5. Click "Stop Stream" when finished.

Applications

- Identity verification
- Interview screening
- Research studies
- Demonstrations

Mode 4: Screen Capture Monitoring

Purpose

Analyze participants during online meetings and interviews.

Procedure

1. Join an online meeting.
2. Open SUDARSHAN AI.
3. Click:

text Detect Meet Window

4. Capture a preview frame.
5. Select the target face.

6. Start monitoring.

Features

- Target-specific tracking
- Continuous monitoring
- Real-time manipulation alerts

Alert Mechanism

When suspicious activity is detected:

text SUSPICION ALERT

is displayed with visual indicators.

8. Understanding Detection Results

Detection Confidence Score

Represents the model's confidence regarding authenticity.

Categories

Confidence	Interpretation
0–30%	Likely Authentic
31–70%	Requires Review
71–100%	Potential Manipulation Detected

Explainability Reports

User Report

Provides simplified explanations suitable for non-technical users.

Developer Report

Provides technical insights into:

- Frequency anomalies
- Temporal inconsistencies
- Spatial irregularities
- Model attention regions

9. Troubleshooting

Issue: Installer Opens and Closes Immediately

Cause

Python is not installed or unavailable in PATH.

Solution

Reinstall Python and ensure:

text Add python.exe to PATH

is checked.

Issue: macOS Blocks Application

Cause

macOS Gatekeeper restrictions.

Solution

1. Right-click the application.
2. Select "Open".
3. Confirm the security prompt.

This procedure is only required once.

Issue: Webcam Cannot Be Accessed

Cause

Another application is using the camera.

Solution

Close applications such as:

- Zoom
- Microsoft Teams
- Google Meet
- OBS Studio

and restart SUDARSHAN AI.

Issue: Tkinter Module Missing on Linux

Error

text ImportError: No module named tkinter

Solution

Run:

bash sudo apt-get install python3-tk

and relaunch the application.

10. Frequently Asked Questions

Is an internet connection required?

No. Core analysis functionality operates locally after installation.

Can the software analyze images?

Yes. Static image analysis is supported.

Can the software analyze live webcam feeds?

Yes. Real-time webcam monitoring is supported.

Can the software monitor online interviews?

Yes. Screen Capture Monitoring mode is designed specifically for interview and meeting environments.

What video formats are supported?

MP4, AVI, and MOV.

What image formats are supported?

JPG, JPEG, PNG, and WEBP.

11. Support

For installation assistance, bug reports, licensing inquiries, or research collaborations, please contact the authorized distribution team.

12. Disclaimer

SUDARSHAN AI is intended solely for academic, research, educational, and authorized evaluation purposes. Detection outcomes represent algorithmic assessments and should not be treated as conclusive evidence without appropriate human review and verification.

Users are advised to comply with all applicable laws, institutional policies, privacy regulations, and licensing terms while using the software.